ELECTROMAGNETIC FILEDS AND WAVES

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C01-3

EXPERIMENT 3

AIM:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Test case (i) Determination and plotting of force between two electric charges with varying distance

PROGRAM:

```
clc;

clear;

q1 = input('Enter the charge q1=');

q2 = input('Enter the charge q2=');

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.1,10,100);

F = k*q1*q2./(r.^2);

disp('Force at r = 1 m is:');

disp(k*q1*q2/(1^2));

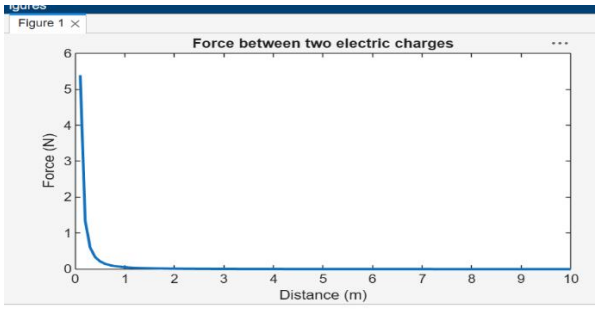
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two electric charges');
```

OUTPUT:



Test case (ii) Determination and plotting of force between two electric charges with different magnitudes

PROGRAM:

```
clc;

q1 = 5e-6;

q2 = 2e-6;

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.5,5,100);

F = k*q1*q2./(r.^2);

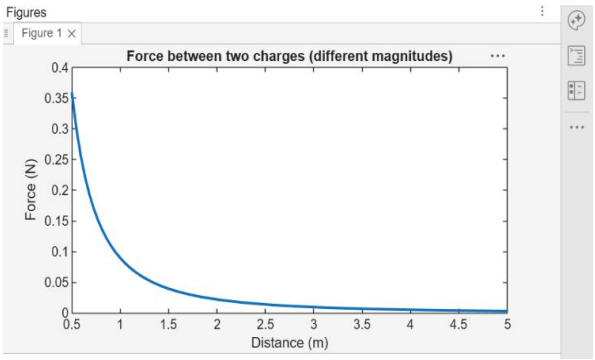
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two charges (different magnitudes)');
```

OUTPUT:



Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance

PROGRAM:

```
clc;

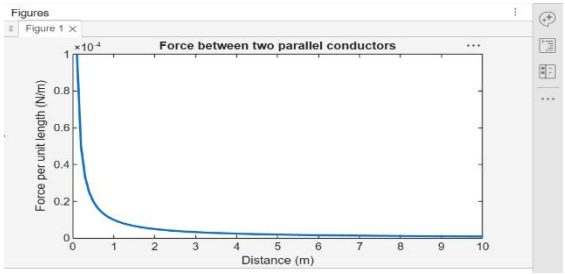
mu0 = 4*pi*1e-7;

I1 = 5;

I2 = 10;
```

```
d = linspace(0.1,10,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between two parallel
conductors');
```

OUTPUT:

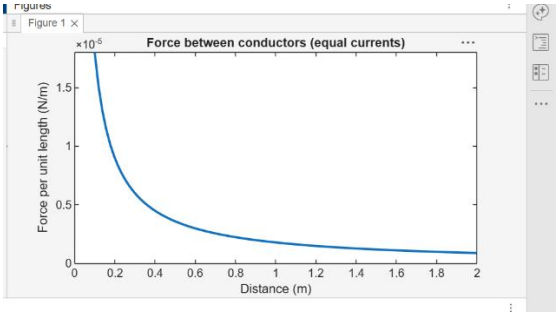


Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

PROGRAM:

```
clc;
mu0 = 4*pi*1e-7;
I1 = 3;
I2 = 3;
d = linspace(0.1,2,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between conductors
(equal currents)');
```

OUTPUT:



RESULTS:

Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using MATLAB was successfully.

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities..	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallelcurrent force calculations correctly with units.	Calculations are mostly correct with minor errors..	Some calculations are correct but require assistance.	Force calculations are invalid or absent	25	2.5	
3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence	Trend and plots are correct with minoromissions.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30	3	
4	Explain and report results	Explain and report results	Explains attraction/repulsi on and	Conclus ion is correct with minor docume	Conclusion is incomplete or partly	Conclusion is missing or incorrect.	20	2	

			reports assumptions, units and conclusions clearly.	ntation gaps.	inaccurate.				
Total							100	10	